CS23334 Fundamentals of Data Science Lab

Experiment 7

Random Sampling and Sampling Distribution

Objective: To explore random sampling from a population and understand the concept of sampling distribution using Python in Jupyter Notebook.

Steps:

1. **Generate a Population:** Create a population of data with a specified distribution (e.g., normal distribution).
2. **Random Sampling:** Perform random sampling from the population to create multiple samples of different sizes. Compute sample statistics (mean, standard deviation, etc.) for each sample.
3. **Sampling Distribution:** Plot histograms or density plots of sample statistics (e.g., sample means). Compare the sampling distribution of the sample statistic (mean) with the population distribution.
4. **Central Limit Theorem (Optional):** Demonstrate the Central Limit Theorem by showing that as sample size increases, the sampling distribution of the sample mean approaches a normal distribution regardless of the population distribution.

Code:

```
import numpy as np
import matplotlib.pyplot as plt

population_mean = 50
population_std = 10
population_size = 100000
population = np.random.normal(population_mean, population_std, population_size)

sample_sizes = [30, 50, 100]
num_samples = 1000
sample_means = {}

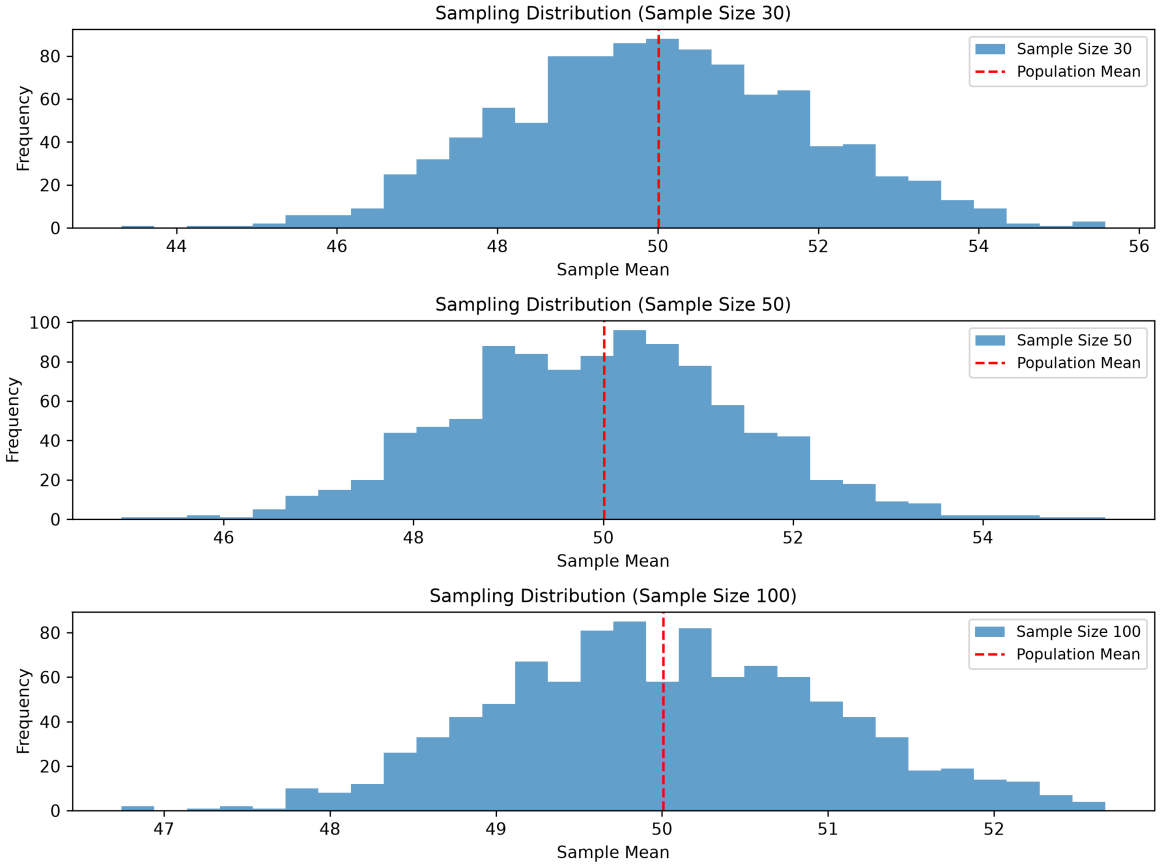
for size in sample_sizes:
    sample_means[size] = []
    for _ in range(num_samples):
        sample = np.random.choice(population, size=size, replace=False)
        sample_means[size].append(np.mean(sample))

plt.figure(figsize=(12, 8))

for i, size in enumerate(sample_sizes):
    plt.subplot(len(sample_sizes), 1, i + 1)
    plt.hist(sample_means[size], bins=30, alpha=0.7, label=f'Sample Size {size}')
    plt.axvline(np.mean(population), color='red', linestyle='dashed', linewidth=1.5, label='Popul
    plt.title(f'Sampling Distribution (Sample Size {size})')
    plt.xlabel('Sample Mean')
    plt.ylabel('Frequency')
    plt.legend()

plt.tight_layout()
plt.show()
```

OUTPUT:



Result:

Random sampling experiments confirmed the Central Limit Theorem — as sample size increased, the sampling distribution of the mean approached a normal distribution. Histograms for sample sizes of 10, 50, and 100 clearly illustrated this convergence.